

Given:

 $A =$ 

$$\begin{bmatrix} 2 & 4 & -3 & -6 \\ 7 & 14 & -6 & -3 \\ -2 & -4 & 1 & -2 \\ 2 & 4 & -2 & -2 \end{bmatrix}$$

RREF Form:

$$\begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

a.) Basis for the column space:

To get the pivot columns, find the leading 1 in the REF or RREF.

Here, the pivot columns are column 1 and 3 in the RREF.

Now, to get a basis for the column space of  $A$ , take the corresponding columns of the original matrix  $A$ .

So a basis is:  $\left\{ \begin{bmatrix} 2 \\ 7 \\ -2 \\ 2 \end{bmatrix}, \begin{bmatrix} -3 \\ -6 \\ 1 \\ -2 \end{bmatrix} \right\}$

b.) Rank of the matrix:

$$\text{Rank} = \text{number of pivot columns} = \text{number of nonzero rows} = 2.$$